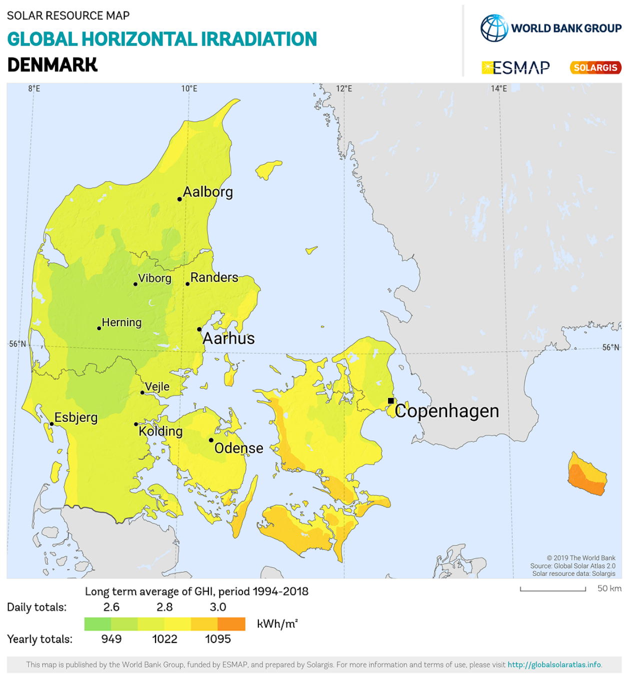
Mass transfer
Modelling 2025

Target for today:

1. Complete a lumped-capacitance heat transfer model
2. Learn two approaches for modeling mass transfer
3. Practice formulating and implementing mass transfer models

8.15-8.20: Warm-up problem
8.20-8:40: Coffee solution
8:40-9:00: Mass transfer lecture
9:00-9:10: Break
9:10-9:30: Mass transfer lecture
9:30-9:40: Menti concept quiz
9:40-10:00: Dissolved oxygen exercise

3. Assuming a forest somewhere in Denmark has a net primary productivity of 5 Mg ha^{1} yr^{-1} on a carbon (C) basis, estimate the net efficiency of solar energy capture.



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Diffusion and Fick's law

Diffusion of mass is similar to heat condution - it is caused by random molecular motion

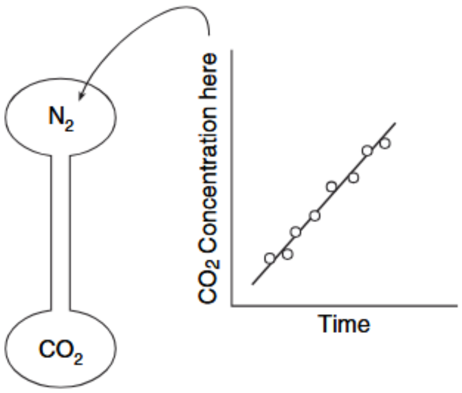
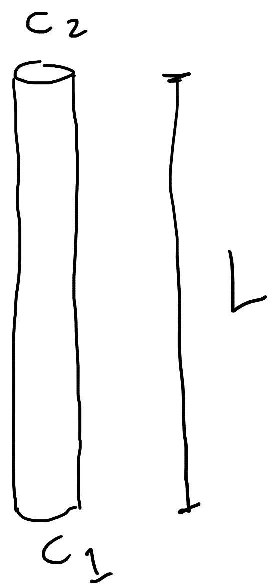


Fig. 1.1-1. A simple diffusion experiment. Two bulbs initially containing different gases are connected with a long thin capillary. The change of concentration in each bulb is a measure of diffusion and can be analyzed in two different ways.

Fundamental constitutive equation is Fick's law

$$J = D \cdot \frac{\Delta c}{L}$$



J = mass flux

e.g. $g\,m^{-2}\,s^{-1}$

c = concentration of diffusing species

e.g. $g\,m^{-3}$

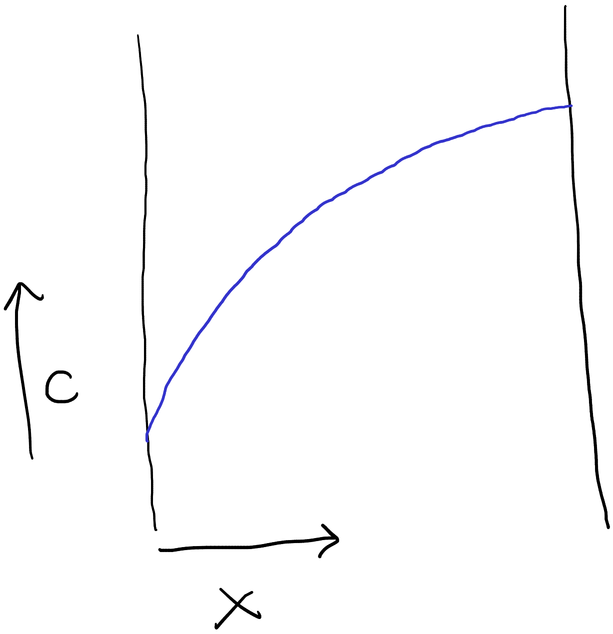
D = diffusion coefficient or diffusivity e.g. $m^2\,s^{-1}$

Fick's law

Like Fourier's law, that form of Fick's law really only applies to linear gradients, which mean steady state with uniform cross-sectional area

A more general form is

$$J = -D_{A,B} \cdot \frac{dc}{dx}$$



But we can often get away with the assumption of a linear gradient

Diffusion coefficients or diffusivity

The diffusion coefficient or diffusivity depends on the diffusing species and the material through which it diffuses!

Table 5.1-1 *Experimental values of diffusion coefficients in gases at one atmosphere*

Gas pair	Temperature (K)	Diffusion coefficient (cm^2/sec)
Air–benzene	298.2	0.096
Air–CH_4	282.0	0.196
Air–C_2H_5OH	273.0	0.102
Air–CO_2	282.0	0.148
Air–H_2	282.0	0.710
Air–H_2O	289.1	0.282

Somewhere around 10^{-1} cm^2/s

$=$ m^2/s

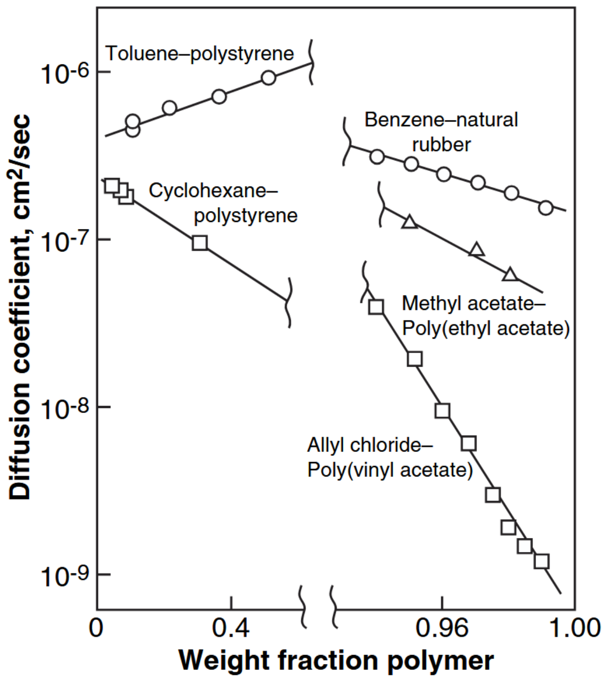
Do you see why Fick's law is a constitutive equation?

Table 5.2-1 *Diffusion coefficients at infinite dilution in water at 25 °C*

Solute	$D(\cdot 10^{-5}\ cm^2/sec)$
Acetic acid	1.21
Acetone	1.16
Ammonia	1.64
Argon	2.00
Benzene	1.02
Benzoic acid	1.00
Bromine	1.18
Carbon dioxide	1.92
Carbon monoxide	2.03

around $10^{-5}\ cm^2/s = 10^{-9}\ m^2/s$

Polymers



For diffusion in gases and liquids, D can be estimated from theoretical and empirical models

For gases, Chapman-Egskog theory, Stokes-Einstein for liquids.

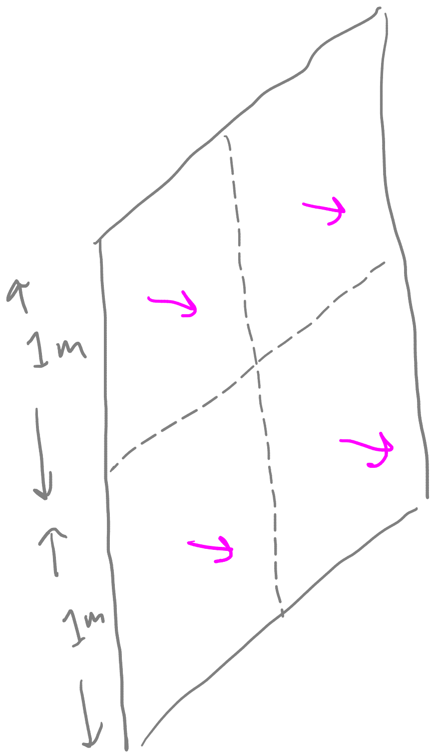
Accuracy around 10-20%

$$D = \frac{1.86 \cdot 10^{-3} T^{3/2} (1/\tilde{M}_1 + 1/\tilde{M}_2)^{1/2}}{p\sigma_{12}^2 \Omega}$$

Diffusivity is much lower for other solids, possibly many orders of magnitude!

Mass flow and mass flux

The concept and math for flux and flow are the same as in heat transfer.



$A = 4\,m^2$

$J = 1\ g\,m^{-2}\,s^{-1}$

$\dot{m} = J \cdot A =$

Lumping transport mechanisms with a mass transfer coefficient

The second constitutive equation we'll deal with does not have a clear name

"mass transfer coefficient approach"?
"convective mass transfer law"?

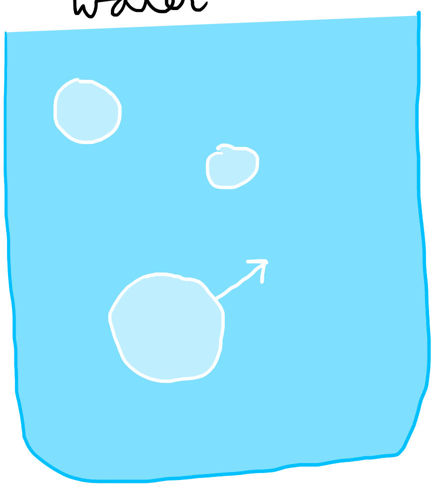
$$J = k_c \cdot \Delta c$$

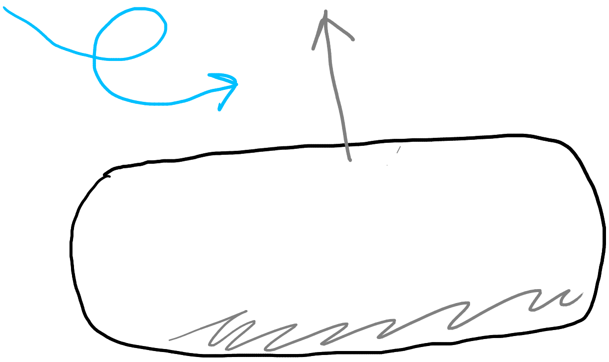
k_c = mass transfer coefficient

e.g. $m\ s^{-1}$

Like convective heat transfer and Newton's law of cooling, it may be applied to the interface between a solid and a fluid, but is often applied at other interfaces as well

O_2 or O_3 into water





Drug dissolution into water

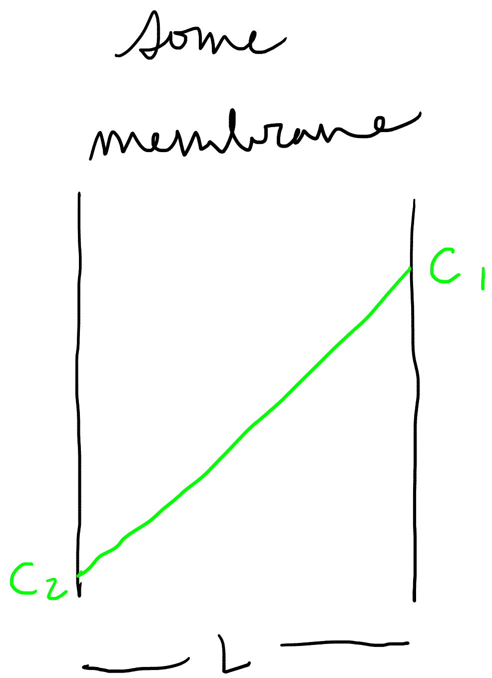
What is special about one of the phases in these two examples?

Equivalence of diffusion and mass transfer coefficient approaches

Just like Fourier's and Newton's laws, do you see the similarity in form?

$$J = D \cdot \frac{\Delta C}{L}$$

$$J = k_c \cdot \Delta C$$



In many cases we could use either approach.

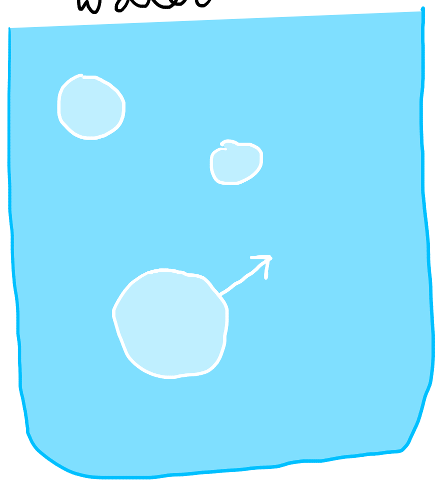
Some advice:
* Use D and diffusion when the profile itself is important or steady-state is not a good assumption
* Use k and the MTC approach when you know that multiple mechanisms are lumped together

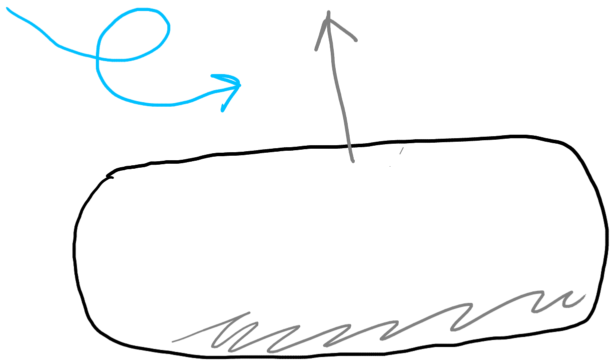
Interphase mass transfer - through an interface

These two examples both have a pure phase!

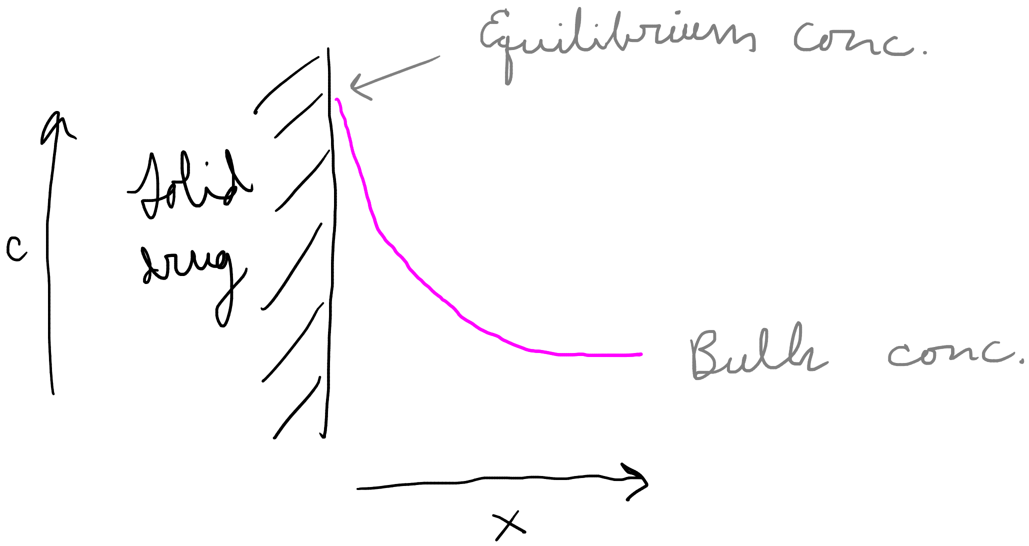
Diffusion does not limit mass transfer in a pure phase

O_2 or O_3 into water



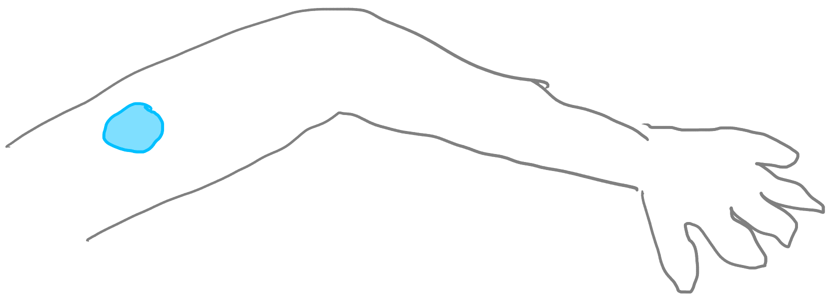


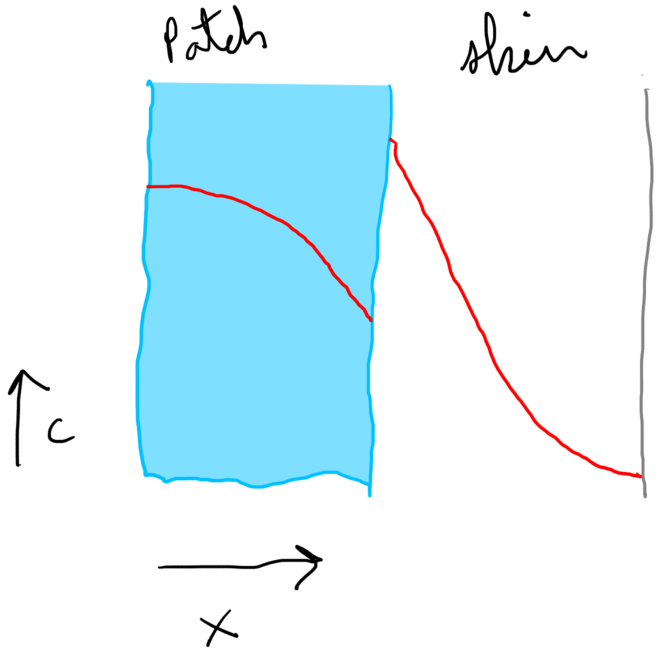
Drug dissolution into water

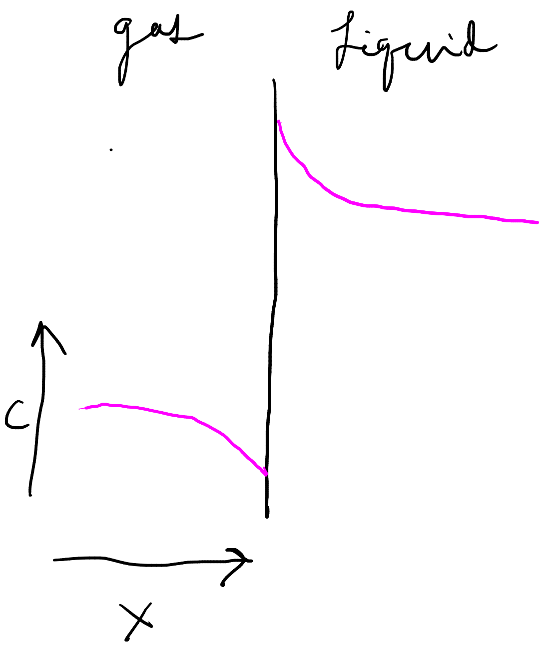


Mass transfer between two mixed phases is more common, and unfortunately more complicated

Interphase mass transfer







Overall mass transfer coefficients

sometimes K (vs. k)

$$J = k_G \, \Delta c_G = k_L \, \Delta c_L$$